

1.2 a)

| Diameter (mm) | Area (cm <sup>2</sup> ) |
|---------------|-------------------------|
| 25            | 4.91                    |
| 50            | 19.64                   |
| 75            | 44.18                   |
| 100           | 78.54                   |
| 125           | 122.72                  |
| 150           | 176.71                  |
| 200           | 314.16                  |
| 300           | 706.86                  |
| 450           | 1590.43                 |

b) Dividing wafer area by die area, approximately 159,043 1mm × 1mm dice.

c) Dividing wafer area by die area,

$$\frac{159,043.13\text{mm}^2}{625\text{mm}^2} = 254.47,$$

so approximately 254 dice.

1.5 Substituting  $Y = 2020$  into the given formula,

$$N(2020) = 1027 \times 10^{0.1505(2020-1970)} = 3.44 \times 10^{10},$$

around 34 billion transistors per microprocessor.

a) Using the given formula,

$$N(Y) = 2N(Y) \Rightarrow Y = \frac{\log_{10}(2)}{0.1505} = 2.002,$$

and thus the transistor count on a microprocessor would be expected to double about every two years.

b) Using the same formula,

$$Y = \frac{\log_{10}(10)}{0.1505} = 6.6445,$$

and thus the transistor count on a microprocessor would be expected to increase by a factor of 10 roughly every six and half years.

1.12 a) Using Fig 1.1(b), we may assume that approximately 175 10 × 10mm dice are on a 150mm wafer. Assuming 35% of the dice are good, this leaves us with 175 \* 0.35 = 61.25, so approximately 61 good dice. Assuming a cost per wafer of \$1,000, the cost per die is approximately

$$\frac{\$1000}{61} = \$16.39.$$

- b) Again using Fig 1.1(b), we may assume that approximately 314  $10 \times 10$ mm dice are on a 200mm wafer. Using the same perecentage of good dice as above, this leaves  $314 * 0.35 = 109.90$ , or approximately 109 good dice. Then, approximate cost per wafer is

$$\frac{\$1000}{109} = \$9.17.$$

1.14 See below for included flowchart:

